

Appendix 1

Supplementary Methods: *Measurement of Elastic Modulus via Impulse Excitation Technique*

The elastic modulus of fused silica was measured using the Impulse Excitation Technique (IET) with an IMCE RFDA HT1600 system. This non-destructive technique enables precise characterization of elastic and damping properties by analysing the vibrational response of a specimen subjected to mechanical excitation. In accordance with ASTM E1876-01, the method provides key mechanical parameters such as Young's modulus (E), shear modulus (G), Poisson's ratio (ν) and internal friction.

Measurement Principle

IET operates by imparting a controlled mechanical impulse to a specimen — commonly in the form of a thin rectangular bar — *via* a striker or automated impactor as shown in Figure S1. The resulting transient vibrations are recorded by a piezoelectric sensor or microphone positioned at an acoustically optimal location. The recorded time-domain response is converted into the frequency domain using Fast Fourier Transform (FFT) analysis to extract the fundamental resonance frequency (f_f). Young's modulus (E) is then calculated using the following expression as standardized by ASTM E1876-01 [S1]:

$$E = 0.9465 \left(\frac{m f_f^2}{w} \right) \left(\frac{L^3}{t^3} \right) T$$

where f_f is the fundamental flexural frequency, m is the mass of the specimen, L , w and t are the specimen's length, width and thickness, respectively, and T is a correction factor that accounts for geometric and boundary condition effects.

Specimen Preparation and Conditions

Specimens were machined into rectangular bars with accurately measured geometries and uniformly polished surfaces to minimize geometric deviations and surface-induced damping. All

measurements were conducted under free-free boundary conditions to reduce constraint-induced frequency shifts. The supports were placed at nodal points to minimize extraneous damping effects and ensure accuracy in frequency measurement.

Environmental Control

IET measurements were performed at both ambient and elevated temperatures (up to 1,000 °C) in a controlled atmospheric environment using the HT1600 furnace attachment. Special care was taken to minimize the influence of environmental variables such as thermal gradients, humidity and ambient gas composition, all of which can affect vibrational characteristics. The accuracy and reproducibility of the IET measurements were ensured by (i) verifying material isotropy and homogeneity, (ii) calibrating sensor sensitivity and striker consistency, (iii) using high-resolution balances and callipers for mass and dimension measurements, and (iv) repeating tests at each temperature and averaging values to account for thermal noise and measurement drift. This methodology facilitated reliable extraction of temperature-dependent Young's modulus values, which were later integrated into the elastoplastic modelling framework described in the main text.

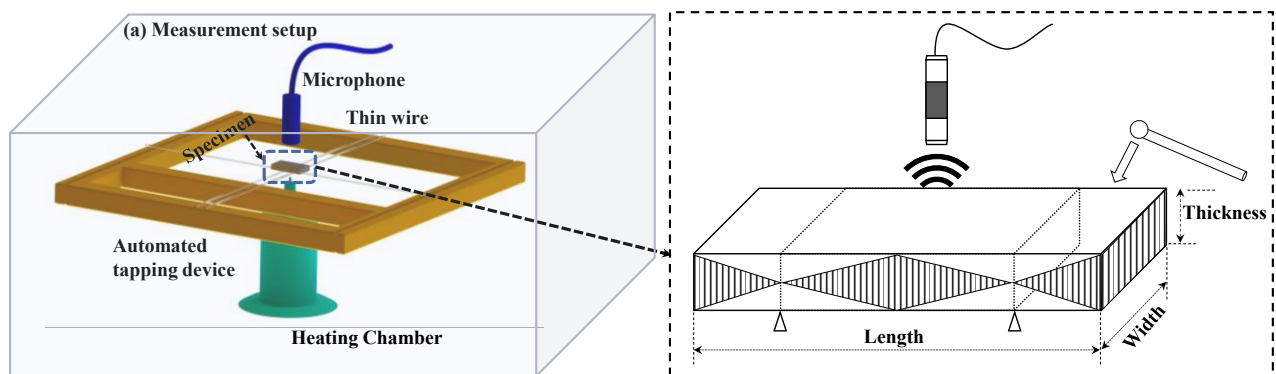


Fig. S1. Schematic diagram of elastic modulus using impulse excitation technique.

Appendix 2

Supplementary Methods: *Measurement of Yield Stress of Fused Silica*

A key challenge in assessing the mechanical response of fused silica is the direct determination of its yield stress, owing to its intrinsic brittleness. Hardness-based estimation provides a practical alternative, as hardness characterizes resistance to localized plastic deformation while yield stress marks the onset of plasticity. For metals, Ashby and Jones [S2] and Tabor [S3] established that yield stress is approximately one-third of hardness, an assumption valid when plastic deformation dominates and elastic effects are negligible.

In brittle materials, however, elastic and plastic strains under indentation are often comparable. For fused silica, elastic deformation is substantial, rendering such empirical correlations inadequate. Instead, brittle solids are more accurately described by an elastic–perfectly plastic framework, in which plasticity occurs immediately upon reaching the yield point without hardening [S4, S5]. Building on this, refined theoretical models have established hardness–yield stress relationships that explicitly account for low elastic moduli [S4, S5]. One such relationship is:

$$\begin{aligned}\frac{H}{\sigma^{YS}} &= \frac{2}{3} \left(1 + \ln \frac{E}{3(1-\nu)\sigma^{YS}} \right) \\ &= \frac{2}{3} (1 - \ln 3(1-\nu)) + \frac{2}{3} \ln \frac{E}{\sigma^{YS}} \\ &= A + B \ln \frac{E}{\sigma^{YS}}\end{aligned}$$

where $A = \frac{2}{3}(1 - \ln 3(1-\nu))$ is a constant depending on Poisson's ratio of materials. According to the value Poisson's ratio of fused silica $\nu = 0.148$ measured by impulse excitation technique, thus $A = 0.041$ and $B = 2/3$ is a constant.

The yield stress was determined by plotting both sides of above Equation, with the intersection of the two curves defining σ^{YS} . As shown in Figure S2, this analysis results $\sigma^{YS} \approx 5.5$ GPa at room temperature and $\sigma^{YS} \approx 3.24$ GPa at 500 °C. Furthermore, a summary of the measured material properties as a function of temperature is presented in Table S1.

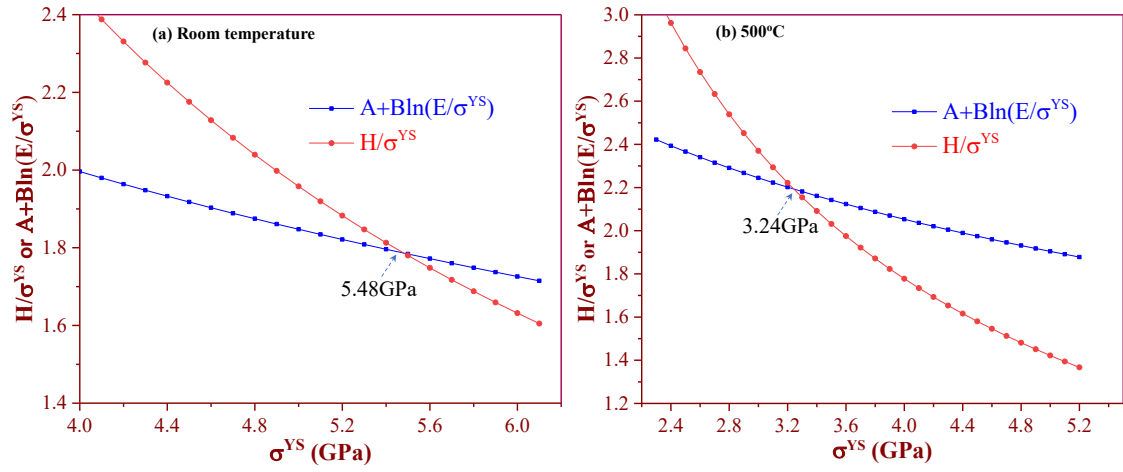


Fig. S2. Determination of yield stress of fused silica.

Table S1. A comprehensive summary of the measured material properties as a function of temperature.

Material properties	Temperature (°C)										
	25	100	200	300	400	500	600	700	800	900	1000
Nanohardness (GPa)	9.8	9.6	8.9	8.5	7.6	7.1	6.6	6.3	5.34	5.0	4.4
Elastic Modulus (GPa)	75.9	77.4	78.9	80.0	81.3	81.8	82.6	83.4	84.2	84.9	86.0
Yield stress (GPa)	5.49	5.21	4.58	4.21	3.60	3.24	2.92	2.72	2.15	1.95	1.6
C value (Ratio of nanohardness and yield stress)	1.79	1.84	1.94	2.01	2.12	2.19	2.27	2.32	2.48	2.6	2.76

Appendix 3

Supplementary Methods: *Densification Characterization of Fused Silica*

The nanoscale structural evolution of fused silica under scratch deformation was characterized using nano-beam electron diffraction (NBED). All measurements were conducted using a Titan ETEM 300 kV D3502 SuperTwin transmission electron microscope (FEG-equipped), operating at an accelerating voltage of 300 kV, an extraction voltage of 3400 V, and an emission current of 242.2 μ A. A small convergence angle of 0.95 mrad and a camera length of 0.100 m were selected to achieve high angular resolution. Diffraction patterns were acquired at 4096×4096 pixels, with analysis focused on the central 1024×1024 region for optimal signal-to-noise.

Error and Uncertainty Analysis

Uncertainty in the density ratio was analyzed, with sources divided into experimental and analytical aspects. Experimental uncertainties include specimen thickness variations (± 5 nm), dark current noise ($\pm 2\%$ intensity variation), and beam damage during multiple measurements. Analytical uncertainties involve statistical errors in RDF peak determination (± 0.01 Å), diffraction pattern fitting quality, and azimuthal averaging. To minimize these, we used 18-frame combinations and repeated measurements (5 times per position) with an optimized exposure time of 1.3 s. The analysis confirms that the observed densification variations are statistically significant, with larger error bars typically corresponding to lower signal-to-noise or thickness variation regions.

Reduced Radial Distribution Function (RDFs) Analysis

The reduced radial distribution functions (RDFs) were computed from azimuthally averaged intensity profiles $I(q)$, extracted from the NBED data. RDF analysis enables the quantitative mapping of local structural changes, including shifts in Si–O bond lengths, modifications in coordination number, and changes in local atomic packing density. These variations serve as direct indicators of densification within the plastically deformed zones.

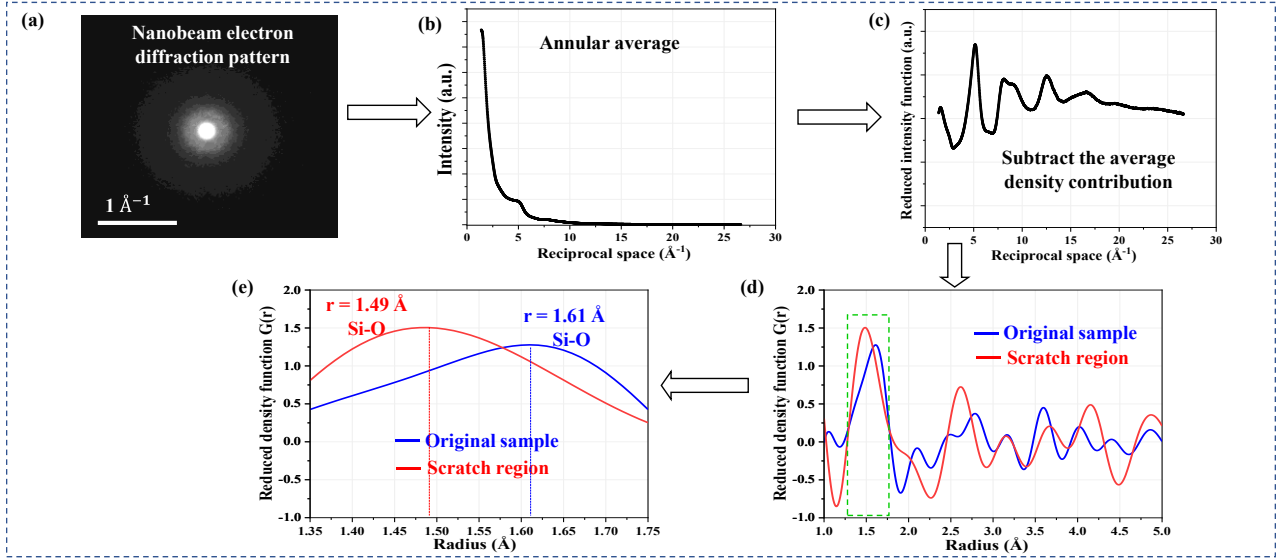


Fig. S3 | Nano-beam electron diffraction (NBED) data processing workflow for densification analysis in fused silica. **a** Representative NBED pattern obtained from a fused silica sample, showing the diffuse rings characteristic of amorphous materials. Scale bar: $1 \text{ \AA}^{-1}$. **b** Azimuthally averaged intensity profile $I(q)$ extracted from the diffraction pattern as a function of reciprocal space vector q . **c** Reduced intensity function $\phi(q)$ calculated after subtracting the average density contribution, emphasizing the structural oscillations. **d** Comparison of reduced density functions (RDFs) between the original undeformed sample (blue) and the scratch-affected region (red), with the green dashed box highlighting the region of interest for Si-O bond analysis. **e** Magnified view of the Si-O bond region ($r = 1.35\text{--}1.75 \text{ \AA}$) showing the shift in bond length from $1.61 \text{ \AA}$ in the original sample to $1.49 \text{ \AA}$ in the scratched region, providing direct evidence of densification-induced structural compression.

The RDF calculation followed Debye's 1915 method, relating diffraction intensity to local atomic density [S6]. The intensity distribution $I(q)$ follows:

$$I(q) = Nf^2 \left\{ 1 + \int_0^R 4\pi r^2 (\rho(r) - \rho_0) \frac{(\sin qr)}{qr} dr + \frac{\rho_0}{q} \int_0^R 4\pi r \sin Kr dr \right\}$$

where q is the scattering vector, N is the number of atoms in the scattering volume, f is the atomic scattering factor, $\rho(r)$ is the local atomic density at distance r from a reference atom, and ρ_0 is the average atomic density. The first term, Nf^2 , represents independent atomic scattering, while the second term, $[\rho(r) - \rho_0]$, accounts for pairwise interference, reflecting local density deviations. The third integral term, describing long-range order effects, theoretically diverges at $q = 0$ but rapidly decays for $q > 0$. In nano-beam electron diffraction, this decay is slower; thus, an optimal q_0 was determined iteratively to ensure its negligible influence, yielding the modified intensity equation as

$$I(q) = Nf^2(q) + 4\pi Nf^2(q) \int_0^\infty [g(r) - \rho_0] \left(\frac{r}{q}\right) \sin(qr) dr$$

where, $g(r)$ denotes the pair distribution function, representing the atomic density probability at distance r . The q -dependence of $f(q)$ captures the angular scattering distribution. The reduced intensity function $\varphi(q)$ was obtained via normalization [S7], i.e.,

$$\varphi(q) = \left[\frac{I(q) - Nf^2(q)}{Nf^2(q)} \right] q$$

In this process, we employed Lobato's parameterization of atomic scattering factors [S8], which provides accurate scattering parameters while maintaining all physical constraints. The reduced density function $G(r)$ was then calculated using the Fourier transform, i.e.,

$$G(r) = 4 \int_0^\infty \varphi(q) \sin(qr) dq$$

Densification was quantified by analyzing Si-O bond length variations. Assuming each SiO₂ molecule as a sphere, density changes were correlated with bond length under constant mass. The density ratio $(r_0/r)^3$ was computed, where r_0 (≈ 1.61 Å) is the initial bond length. Diffraction data were processed using our custom MATLAB code.

Table S2. Densification data and measurement uncertainties at different locations under various thermal conditions.

Part 1: Line ① (Red)

Distance from surface (nm)	Mean RT density ratio (ρ/ρ_0)	Standard deviation	Mean 400°C density ratio (ρ/ρ_0)	Standard deviation	Mean 800°C density ratio (ρ/ρ_0)	Standard deviation
50	1.115469	0.020746	1.107136	0.073701	1.153932	0.01302
100	1.004952	0.023811	1.079576	0.11853	1.12759	0.057584
150	1.023946	0.024414	1.023915	0.103718	1.137867	0.041047
200	1.048382	0.025618	0.996525	0.04763	1.02854	0.011169
250	0.981589	0.091359	1.033869	0.033535	1.05805	0
300	1.000579	0.034037	0.979295	0.067563	1.018987	0.015601
350	0.991482	0.038348	0.99575	0.028308	1.033619	0.024285
400	0.993857	0.076361	1.005517	0.040635	1.009549	0.018636
450	1.019222	0.026691	1.003432	0.13097	1.024066	0.028945
500	0.948137	0.062049	1.035154	0.060126	0.973742	0.029107
550	1.00197	0.062884	1.024933	0.051841	1.006321	0.056886
600	0.976629	0.086218	0.968564	0.030254	1.043422	0.025036
650	1.028783	0.024982	1.015874	0.058652	0.987088	0.042512
700	1.02854	0.011169	1.019943	0.046866	1.043295	0.018779
750	0.98415	0.070357	1.006245	0.056533	0.933823	0.068314
800	1.024316	0.037135	1.029739	0.050519	1.005068	0.028231
850	1.023704	0.009673	0.998521	0.081859	0.939434	0.010893
900	1.009669	0.024365	1.024779	0.047708	1.000579	0.034037
950	1.025359	0.058317	1.006321	0.056886	0.964058	0.028308
1000	1.028663	0.019591	1.002897	0.076209	1.023823	0.018314

Part 2: Line ② (Yellow)

Distance from surface (nm)	Mean RT density ratio (ρ/ρ_0)	Standard deviation	Mean 400°C density ratio (ρ/ρ_0)	Standard deviation	Mean 800°C density ratio (ρ/ρ_0)	Standard deviation
50	1.141422	0.096207	1.142531	0.158058	1.140501	0.086061
100	1.162083	0.056084	1.178971	0.074496	1.171144	0.021904
150	1.142523	0.045789	1.144509	0.064495	1.177086	0.029895
200	1.147523	0.102738	1.082212	0.090933	1.160013	0.033936
250	1.097677	0.146089	1.125092	0.096285	1.150297	0.067919
300	1.107958	0.143409	1.150297	0.067919	1.149993	0.062259
350	1.034575	0.050055	1.128501	0.074726	1.15957	0.011276
400	1.001049	0.046063	1.177086	0.029895	1.043422	0.025036

450	1.029997	0.055956	1.002212	0.066815	1.028663	0.019591
500	0.983991	0.069444	1.016041	0.062526	1.018987	0.015601
550	0.993381	0.073887	1.010846	0.054573	1.029018	0.032969
600	1.025108	0.053483	1.01717	0.077839	1.016662	0.070178
650	1.010375	0.044945	0.981929	0.0254	0.953295	0.018779
700	1.002897	0.076209	0.994147	0.08299	0.978962	0.0601
750	1.02491	0.0507	0.998146	0.074387	1.02854	0.011169
800	1.014628	0.032704	0.987752	0.055499	1.018987	0.015601
850	1.046135	0.081303	0.998735	0.084175	0.964733	0.035262
900	1.000351	0.026691	1.005627	0.105667	1.018867	0
950	1.076977	0.091971	0.981929	0.0254	1.033869	0.033535
1000	0.986309	0.017426	0.995743	0.02769	1.023946	0.024414

Part 3: Line ③ (Yellow)

Distance from surface (nm)	Mean RT density ratio (ρ/ρ_0)	Standard deviation	Mean 400°C density ratio (ρ/ρ_0)	Standard deviation	Mean 800°C density ratio (ρ/ρ_0)	Standard deviation
50	1.12364	0.138142	1.146231	0.094298	1.133705	0.066939
100	1.142783	0.13884	1.165799	0.039072	1.172779	0.04968
150	1.135174	0.091668	1.181856	0.058196	1.137379	0.023888
200	1.153096	0.176535	1.071615	0.011747	1.186033	0.042076
250	1.163532	0.192979	1.201961	0.197518	1.156676	0.076956
300	1.049717	0.162678	1.158437	0.148093	1.137615	0.033006
350	1.078203	0.145713	1.13647	0.05524	1.126165	0.009515
400	1.024066	0.028945	1.03887	0.039203	1.115767	0.032162
450	0.984404	0.074343	1.099992	0.042307	1.106514	0.064547
500	1.02391	0.10372	1.034093	0.075551	1.033466	0.016226
550	1.025161	0.055845	1.034788	0.153577	1.03841	0.019592
600	1.019715	0.041976	0.994483	0.038921	1.033466	0.016226
650	1.008033	0.130268	1.065867	0.051356	0.996407	0.044463
700	1.023152	0.135861	1.019027	0.019107	1.03366	0.025208
750	0.955965	0.047065	1.038377	0.019592	1.043163	0.00859
800	1.045146	0.064714	0.985907	0.099617	1.000994	0.042442
850	1.039064	0.041677	1.052116	0.041965	1.03841	0.019592
900	0.986993	0.04086	1.001816	0.063041	1.009517	0.016547
950	1.014856	0.03882	1.02097	0.070204	1.053183	0.016641
1000	0.996927	0.05683	0.982943	0.054608	0.96925	0.027036

Part 4: Line ④ (Green)

Distance from surface (nm)	Mean RT density ratio (ρ/ρ_0)	Standard deviation	Mean 400°C density ratio (ρ/ρ_0)	Standard deviation	Mean 800°C density ratio (ρ/ρ_0)	Standard deviation
50	1.079269	0.161617	1.133111	0.059085	1.128749	0.035863
100	1.146602	0.093384	1.153932	0.01302	1.145391	0.078871
150	1.13759	0.057584	1.148872	0.037969	1.164758	0.020267
200	1.133816	0.090741	1.153932	0.01302	1.160461	0.046946
250	1.164937	0.096422	1.132229	0.03743	1.109166	0.04609
300	1.091236	0.073404	1.154226	0.029124	1.172504	0.060662
350	1.147008	0.116028	1.166403	0.052793	1.151382	0.039623
400	1.116011	0.040005	1.177968	0.053649	1.137347	0.066508
450	1.131232	0.107172	1.154226	0.029124	1.145499	0.115728
500	1.07958	0.11853	1.095972	0.066633	1.166702	0.059076
550	1.045717	0.072368	1.12702	0.044911	1.148732	0.03374
600	1.07164	0.087357	1.115889	0.037136	1.130518	0.03648
650	1.04199	0.08788	1.049635	0.057383	1.101032	0.062302
700	1.005822	0.119585	1.043422	0.025036	1.069406	0.05221
750	1.010216	0.053411	1.010016	0.035782	1.074751	0.056775
800	1.024991	0.185662	1.012681	0.08098	1.01619	0.0656
850	1.005068	0.028231	0.983077	0.054537	1.041985	0.087878
900	0.981708	0.014844	1.068351	0.020092	0.99184	0.047202
950	1.045459	0.068331	1.005415	0.038574	1.001035	0.045306
1000	1.017821	0.08881	1.014505	0.028387	1.005636	0.043575

Table note: Density ratio ρ/ρ_0 represents local density relative to undeformed fused silica ($\rho_0 = 2.20 \text{ g/cm}^3$). Standard deviations calculated from four repeated measurements. Locations ①-④ correspond to scan positions marked in Fig. 5(a-c).

References

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